

Introduction

In this report, I go through the steps for property valuation using different modeling techniques. First, I check if a time adjustment is needed by looking at the Sale Price to Assessed Value Ratio (SAR). Then, I calculate land value by finding the land residual, making adjustments for lot size, and using the unit valuation method. I also test if different property features, like cul-de-sacs and alleys, affect land value and make adjustments if needed.

For improvement valuation, I use the Replacement Cost New (RCN) method and apply a market-based depreciation factor. I check if the adjusted values are consistent across property characteristics like split-level designs, garage types, and neighborhood differences. After making necessary adjustments, I compare the final cost value with sale prices to see how well the model performs. The goal is to ensure fair and consistent valuations by using statistical tests and making refinements where needed.

Modeling Steps

Time Adjustment

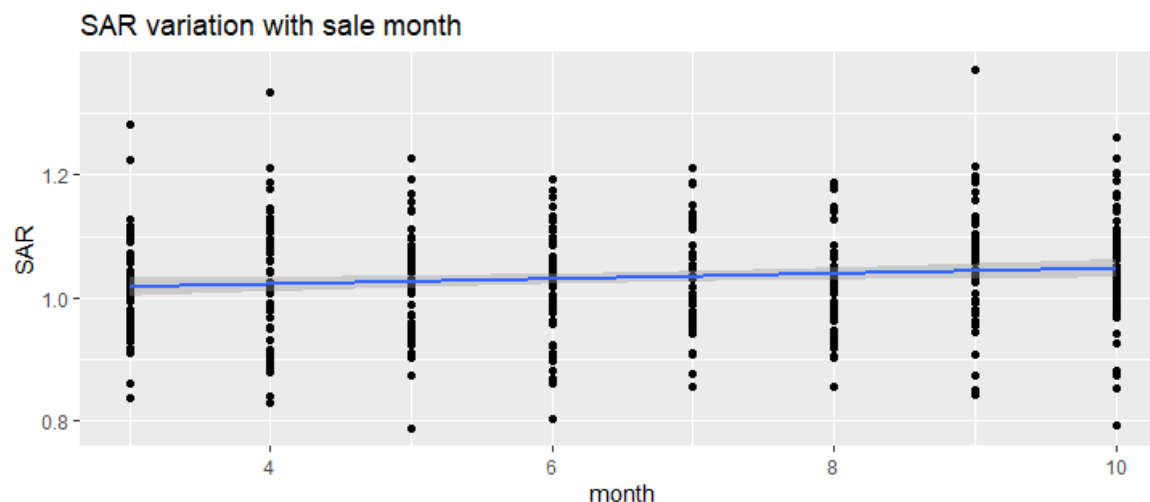
Time Adjusting Sale Price

First, I am using the scale to assessment ration method to evaluate if time adjustment would be required. Our valuation date is July 1st, 2017, and the sales in this database occurred between March 2017 and October, 2017.

Ratio Statistics for Price / Total Assessed Value (previous year)

Mean		1.03
95% Confidence Interval for Mean	Lower Bound	1.02
	Upper Bound	1.04
Median		1.04
95% Confidence Interval for Median	Lower Bound	1.02
	Upper Bound	1.04
	Actual Coverage	95.3%
Minimum		0.79
Maximum		1.37
Coefficient of Dispersion		6.72%
Coefficient of Variation	Mean Centered	8.57%

The 672% COD indicates that the have been acceptable in relation to the sale prices. That is to say, Sale Price to Assessed Value ratio (SAR) ratio can be used for time adjustment. The scatter plot between SAR value and sale month is given below:



Interestingly, there is a continuous increase in the SAR over the sale period. The correlation, however, is not significant as it is indicated by the value for $R^2 = 0.012$. Therefore, it would be more appropriate to apply month adjustment to each month individually, using a non-parametric approach. Here, I use Kruskal-Wallis test for this purpose: the test determines if there is a significant difference between how SAR ranks are distributed between months.

Rank

SAR	Sale Month	N	Mean Rank
	3	56	196
	4	62	222
	5	51	221
	6	66	208
	7	51	237
	8	50	198
	9	48	254
	10	62	256
	Total	446	

Test Statistics: Kruskal-Wallis – Grouping by Sale Month

	SAR
Chi-Square	12.674
df	7
Asymp. Sig	0.08

The asymptotic significance statistic is 8%, which would not small enough to reject the null hypothesis that the SAR values are equally distributed among months. That is to say, time adjustment is not necessary as the SAR rank between months are not significantly different. Note that this observation is also aligned with low value of R^2 , derived before.

Land Value Evaluation

Calculating Land Residual

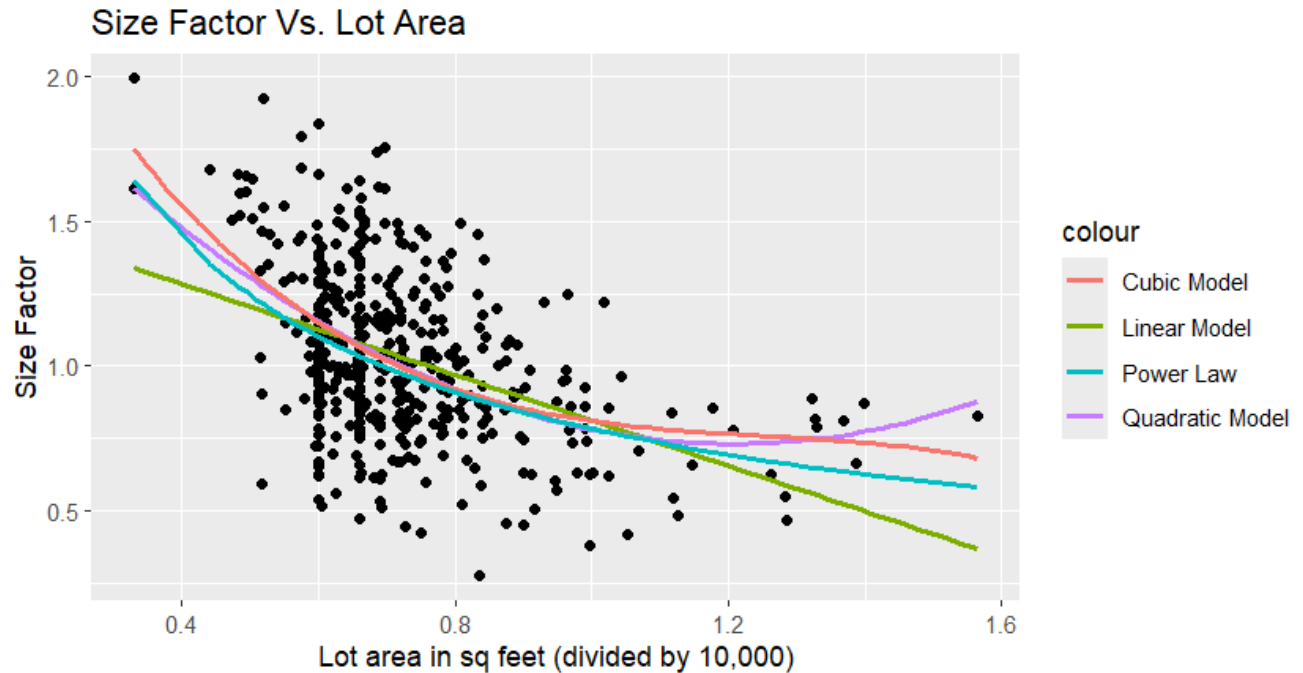
I can find the land residual by subtracting time adjusted sale price and time adjusted improved value. Land residual will then be used as the dependent variable for further analysis. First, I check the land residuals to see if there is any strange or outliers. The variation of land residuals with lot size is shown in Figure A.1 in Appendix. The figure shows that the land residuals value increase as lot sizes increase, which is expected. There is also no negative value for the land residual which is good.

Next, I will check if the land residual per square feet of lot size decreases as the lot increases (Land Value theory). As it is shown by the Figure A.2 in Appendix, the value per square fee of lot size decreases as the lot size increase; this is aligned with our expectation. Next, I should make a size adjustment factor so the relationship between the land residual per square feet and the lot size becomes more linear. This will help developing the model later.

Ratio Statistics for Land Residual / Lot Area

Mean		8.08
95% Confidence Interval for Mean	Lower Bound	7.87
	Upper Bound	8.29
Median		7.83
95% Confidence Interval for Median	Lower Bound	7.58
	Upper Bound	8.06
	Actual Coverage	95.3%
Minimum		2.18
Maximum		15.58
Coefficient of Dispersion		23.17%
Coefficient of Variation	Mean Centered	28.34%

Taking the median = 7.83 as the central tendency, I will then compute the size factor as the ratio between the residual land per lot area and 7.83. Figure below shows the relationship between size factor and the lot area ins sq feet; I divide lot size by 10,000 so the model coefficients' values are in reasonable magnitude.



Comparison between Various Models for Size Factor Equations

Equation	R Square	Adjusted R Square	F	Df1	Df2
Linear	0.18	0.18	98.86	1	444
Quadratic	0.22	0.21	61.96	2	443
Cubic	0.22	0.22	42.08	3	442
Power	0.20	0.20	110.80	1	444

Based on the figure and table above, power model seems to be the appropriate model for size factor. Here are the coefficients for the power model:

$$\text{Size Factor} = a \times \left(\frac{\text{lot size}}{10000} \right)^b$$

$$a = 0.78 \mid b = -0.67$$

Finally, I make another column in the data for the adjusted lot size as follows:

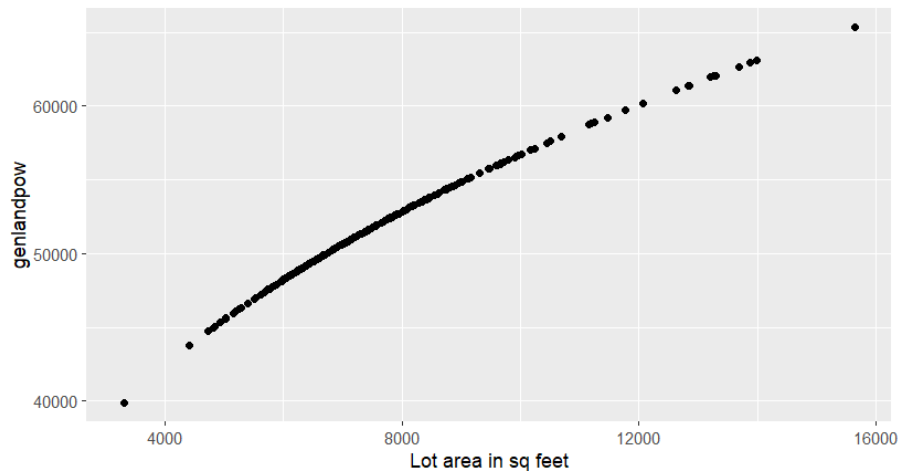
$$\text{Adjusted Lot Size} = \text{Lot Size} \times \text{Size Factor}$$

Now, I can use the adjusted lot size for the modelling. I am using unit value method for land valuation.

Unit Valuation Method

Following the textbook, I define another column for general land value as follows:

$$\text{Generalized Land Residuals Power Law} = \text{Lot Size} \times 7.83 \times \text{Size Factor by Power Law}$$



Note that 7.83 is the median of land residual per square feet. The general land value above give us an estimate of the land residual to work on. To ensure valuation is equal across various land characteristics, I run Mann-Whitney test to binary land characteristics.

Mann-Whitney Results – general land value				
Variable	Value	Number	Mean Rank	Ays. Significance
corner	0	378	220	0.20
	1	68	242	
culdesac	0	417	219	0.00
	1	29	285	
alley	0	301	249	0.00
	1	145	171	
wooded	0	410	225	0.42
	1	36	207	
pie_shpe	0	420	219	0.00
	1	26	293	
slope	0	389	225	0.55
	1	57	214	
abv_road	0	367	228	0.11
	1	79	202	

The table above shows that the general land value may not be equally distributed between various groups. In particular, the value is not equally distributed with respect to `culdesac`, `alley`, and `pie\_shpe` with 95% significance level. Culdesac, pie_shpe and alley show to have a strong correlation with each other; that is expected since lots with culdesac are more prone to be pie_shape and they most likely do not have alley. However, adjustments are likely needed for both culdesac and alley as there is only one lot with both alley and culdesac features.

Lots with Culesac tends to have pi_shpe lots with no Alley

	Alley		pi_shpe	
Culdesac	0	1	0	1
0	273	144	404	13
1	28	1	16	13
	Chi-squared p.value = 0.00		Chi-squared p.value = 0.00	

Ratio Statistics for landresid/gelandpow

Culdesac	Alley	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
0	0	1.01	0.97	1.05	95.52%	0.37	1.76	20.51
1	0	1.20	1.06	1.26	95.72%	0.18	1.58	17.24

Table above suggests adjusting general land value for lots with culdesac by (1.20), those without culdesac but with alley by 0.92, and those with neither by 1.10. I repeat the Mann-Whitney tests, and the results (Table A.1) shows that the general land value still needs to be adjusted for alley. Let's find the ratio statistics for land residual and adjusted general land value for Alley.

Alley	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
0	1.07	1.02	1.10	94.26%	0.15	1.75	20.34
1	0.91	0.87	0.95	94.34%	0.37	1.57	19.41

Similarly, the general land value for lots with alley needs to be adjusted by 0.91 and those without alley by 1.07. Doing the same for abv_road, I adjust land value with 0.94 for those with abv_road and finally another adjustment for those with culdesac (1.01) and those without culdesac (0.96) (Corresponding tables are in Appendix)

Which confirms no more adjustments are needed. Finally, I should check if the ratio is equally distributed over the neighbourhoods. Chi squared test shows that there is a significant difference between the neighborhood and further adjustment is needed.

Neigh	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
713	1.00	0.93	1.04	95.78	0.43	1.52	17.86
715	1.00	0.85	1.06	94.81	0.5	1.73	20.68
716	1.00	0.96	1.04	95.26	0.47	1.46	15.47
731	1.08	0.98	1.29	95.72	0.63	1.67	19.64
732	1.00	0.73	1.41	96.09	0.73	1.49	20.32
734	0.99	0.83	1.07	90.77	0.66	1.59	16.96
736	1.00	0.94	1.11	96.43	0.2	1.49	20.19
737	1.03	0.98	1.11	95.78	0.37	1.62	18.07

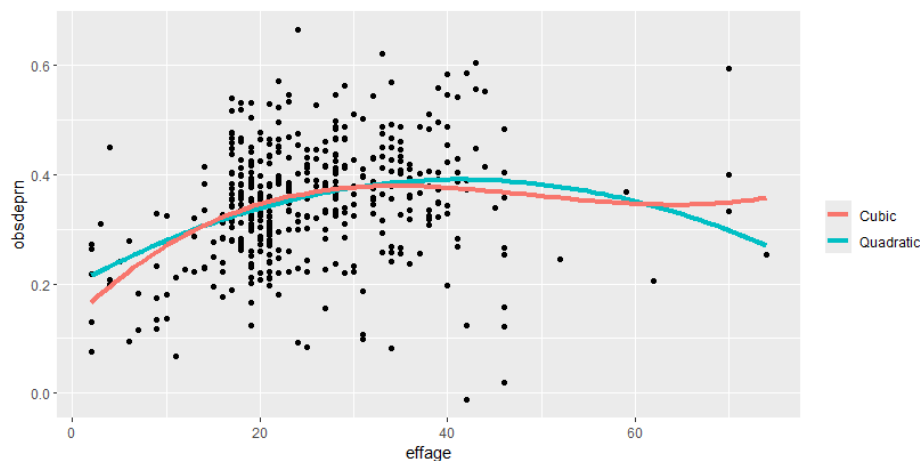
I adjusted the ratio for all neighbourhoods, except 731 and 737 and following is the results. Distribution of land residual ratio after adjustment is available in Appendix (Fig A.2).

Cost Model

Market-based Depreciation Factor

I showed in the previous sections that time adjustments are not needed for sale price and so the improvement value is obtained by subtracting land value calculated from the previous section from the sale price. The obtained improvement value is then divided by sum of all replacement cost new components (i.e., main building, garage, carport, and outbuilding) to results observed depreciation. First, I draw depreciation rate versus effective year to find a model for depreciation.

Observed depreciation over effective year



The figure and table above both show that the quadratic model is modelling the depreciation better than cubic transformation. The model shows a decay in depreciation for the effective years above 40 years old which does not seem logical and hence I create another column (effective year 2) which caps effective year at 40 years old. There is also one depreciation less than 0 that needs to be removed. I refit the model after removing the one with negative depreciation and list the coefficients below:

Parameter Estimates

Equation	Constant	b1	b2
Quadratic	0.351209	0.564770	-0.497089

The table and figure show that the quadratic curve provides a better fit and the curve looks reasonable. Using the quadratic curve, the depreciation can be found as:

$$\text{Depreciation} = 0.350393 + 0.632931 \times \text{effage2} - 0.436044 \times (\text{effage2})^2$$

$$\text{Percentage Good} = 1 - (0.350393 + 0.632931 \times \text{effage2} - 0.436044 \times (\text{effage2})^2)$$

Where percentage good is a value between 0-1 where 1 means no depreciation and 0 is full depreciation. The Percentage Good versus Effective Age figure is available in the Appendix (Fig A.3)

RCN Calculation

I can find the Replacement Cost New (RCN) by summing the estimated costs of the main building, garage, and carport. Each component is calculated separately based on the new cost table. First, I determine whether the building is one or two storeys and whether it has a full basement or a crawl space. A full basement is identified if the finished basement area (BMST_AREA) is greater than zero, while a crawl space is identified if the crawl space area (CRWLAREA) is greater than zero. If the second-floor area (FLRAREA2) is greater than zero, the building is considered two-storey.

Next, I apply the base constant costs. For one-storey buildings, the base cost is \$51,500 if there is a full basement and \$45,500 if there is a crawl space. For two-storey buildings, an additional \$16,300 is added to the base cost. I then apply area-based costs. The first-floor area is multiplied by \$42.50 per square foot, and if there is a second floor, the second-floor area is multiplied by \$33.40 per square foot. Additional costs are added for specific features. If the building has a hot-water heating system (HEATTYPE = 7 or 8), an extra \$4.40 per square

foot for the first floor and \$3.30 per square foot for the second floor is included. Fireplaces add \$3,850 each, and bathrooms add \$4,000 each. Once the total raw cost for the main building is calculated, it is adjusted by the main building class factor. For example, if the building class (MBMANCLS) is in the range of 40 to 42, the factor is 0.507; if it is in the range of 140 to 142, the factor is 1.000. The raw cost is multiplied by this factor to get the final main building RCN. Next, I calculate the garage and carport costs separately. Garage costs are determined based on garage type and area. A detached garage has a base cost of \$16,300 plus \$30 per square foot, while an attached garage has a base cost of \$11,600 plus \$30 per square foot. If the garage is located in the basement, a fixed cost of \$7,100 is applied. This raw cost is then adjusted using the garage quality factor, where lower-quality garages have a factor of 0.450, and higher-quality garages have a factor of 1.420. For carports, the base cost is \$5,850 plus \$22 per square foot of carport area. The final carport RCN is obtained by applying the carport quality factor, similar to garages.

Finally, the total RCN is found by summing the RCN values for the main building, garage, and carport. The calculated RCN value is then multiplied by depreciation (percent good) to find the improved residual values. Next, I am comparing the derived improved value with improved residual value.

Developing Improvement Value

Ratio Statistics for imprrsid / imprcnld

Mean		1.29
95% Confidence Interval for Mean	Lower Bound	1.27
	Upper Bound	1.31
Median		1.29
95% Confidence Interval for Median	Lower Bound	1.26
	Upper Bound	1.31
	Actual Coverage	94.79%
Minimum		0.52
Maximum		2.29
Coefficient of Dispersion		13.91

The table above shows the imprcnld should be adjusted by 1.29 (median) to be better comparable with imprrsid. Therefore,

$$\text{Adjusted Depreciated RCN} = 1.29 \times \text{RCN} \times \text{Percentage Good}$$

The ratio statistics changes as follows, after adjustment

Ratio Statistics for imprrsid / imprcnld_adj

Mean		1.00
95% Confidence Interval for Mean	Lower Bound	0.99
	Upper Bound	1.02
Median		1.00
95% Confidence Interval for Median	Lower Bound	0.98
	Upper Bound	1.02
	Actual Coverage	94.79%
Minimum		0.41
Maximum		1.78
Coefficient of Dispersion		13.91

Testing Building Characteristics

First, I test the adjusted ratio for various building stories. Using the project description, the last digit of main building manual class indicates the number of stories. I define

$$\text{mkfact1} = \text{imprrsid} / \text{imprcnld_adj}$$

Ranking for Mkfact1 versus Stories

Mkfact1	storey	Number	Mean Rank
	0	411	223
	1	12	277
	2	15	213
	5	4	237
	7	3	73.7

Kruskal-Wallis test for Mkfact1 versus Stories

	Mkfact1
Kruskal-Wallis H	2.94
df	2
Asymp. Sig	0.23

Note than I merged those folios with 5 or more stories with 2 stories to ensure I do have enough folios in each group. The results of Kruskal-Wallis test shows that there does not seem to be a significant difference between the mkfact1 between stories. Next, I am repeating the same process with quality classes.

Ranking for Mkfact1 versus Quality Class

Mkfact1	Quality Class	Number	Mean Rank
	4	8	284
	5	3	79
	8	45	228
	9	215	201
	14	168	246
	15	6	315

Kruskal-Wallis test for Mkfact1 versus Quality Class

Mkfact1	
Kruskal-Wallis H	6.33
df	4
Asymp. Sig	0.18

Likewise, I merged the quality classes 4, 5, and 15 to ensure there are enough number of folios in all groups. Kruskal-Wallis shows that the mkfactr1 is equally distributed among the quality classes too. Next, I check the split-level factor for equal distribution of mkfact1.

Ranking for Mkfact1 versus Split-level

Mkfact1	Split Level	Number	Mean Rank
	0	411	219
	1	34	277

Kruskal-Wallis test for Mkfact1 versus Split-level

Mkfact1	
Kruskal-Wallis H	6.52
df	1
Asymp. Sig	0.01

Ratio Statistics for Mkfact1 based on Split-level

Split-level	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
0	0.99	0.97	1.01	95.54	0.41	1.78	13.82
1	1.09	0.97	1.18	95.90	0.67	1.32	12.53

Kruskal-Wallis test shows there is a significant difference between market factors of properties with and without split. I adjust the market improvement RCN using the median of market factor for each group (Mkfact2). Table A.4 shows the adjustment is done correctly.

Repeating the same for garage type, I find out that an adjustment for garage type is needed too as the Kruskal-Wallis test shows significance below 5%. The new market factor is named Mkfact3.

Ratio Statistics for Mkfact2 (adjusted by Split-level) based on Garage Type

Garage Type	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
0	1.01	0.98	1.03	94.4	0.41	1.79	13.51
1	0.99	0.94	1.04	95.64	0.73	1.33	11.45
2	0.89	0.84	0.94	93.73	0.58	1.37	14.86
3	1.09	1.02	1.13	96.49	0.79	1.42	9.98

Table A.5, Table A.6, and Table A.7 shows the ratio statistics after adjustments above for number of bedrooms, carport, and family room and they show more adjustments are needed. I run few more tests to ensure no further adjustments are needed.

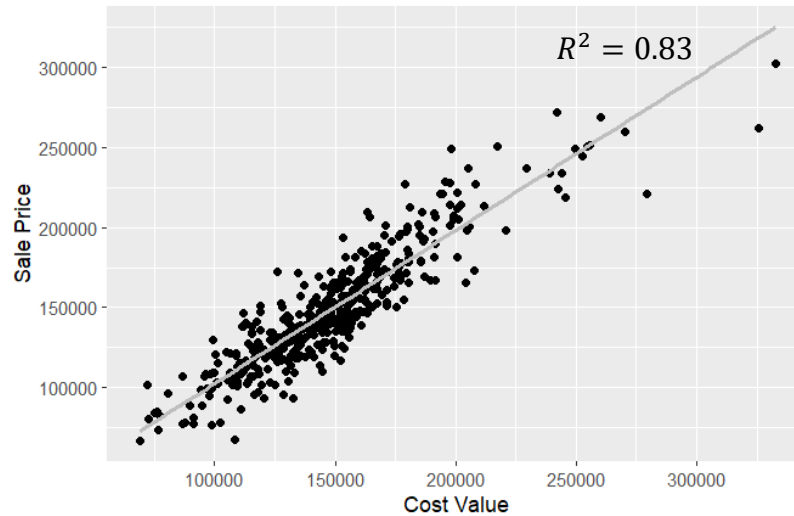
Variable	Kruskal-Wallis Test p_value
Carport	0.32
Fireplaces	0.26
Bedrooms	0.94
Baths	0.42
Basement	0.63
Garage Manual Class	0.33
Carport Manual Class	0.27
Heat Type	0.05

Final Analysis and Conclusion

Model Calibration and Testing

Final Cost Value is obtained by adding the land value found by unit valuation method with the adjusted depreciated RCN I derived above. Comparing it with the sale price

Sale Price Vs. Cost Value



Judging based on the R^2 , there is a very good relationship between Sale Price and Cost Value. Table below shows the ratio statistics for the ratio between Cost Value and Sale Price.

Ratio Statistics for ASR (Cost Value / Sale Price)

Mean	1.01
95% Confidence Interval for	Lower Bound
Mean	Upper Bound
Median	1.00
95% Confidence Interval for	Lower Bound
Median	Upper Bound
	Actual Coverage
Minimum	0.71
Maximum	1.6
Coefficient of Dispersion	8.01

Neighborhood Analysis

Neigh	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
713	1.01	1	1.04	0.71	1.3	95.78	7.76
715	1.02	1	1.04	0.8	1.31	94.81	7.44
716	1	0.99	1.02	0.79	1.26	95.26	6.32
731	0.89	0.83	0.91	0.73	1.09	93.86	8.46
732	1	0.78	1.06	0.78	1.07	96.09	7.41
734	1.22	1.17	1.3	0.99	1.6	90.77	9.57
736	0.97	0.93	1.02	0.83	1.13	96.43	6.87
737	0.97	0.95	1	0.78	1.3	95.78	7.46

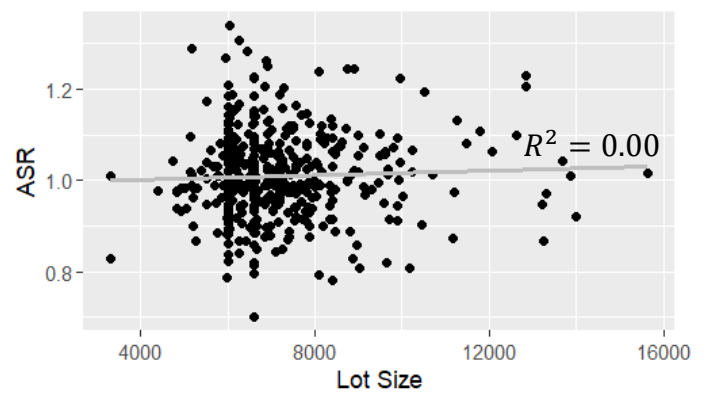
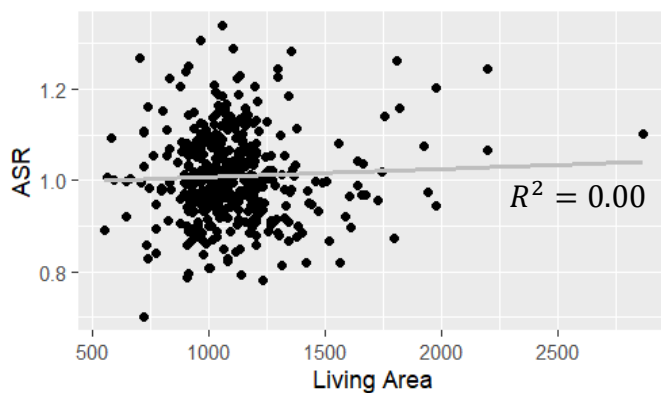
ASR	
Kruskal-Wallis H	65.73
df	7
Asymp. Sig	0.00

Which shows an adjustment is needed for the neighborhood. After adjustment, I review the ratio statistics for adjusted ASR:

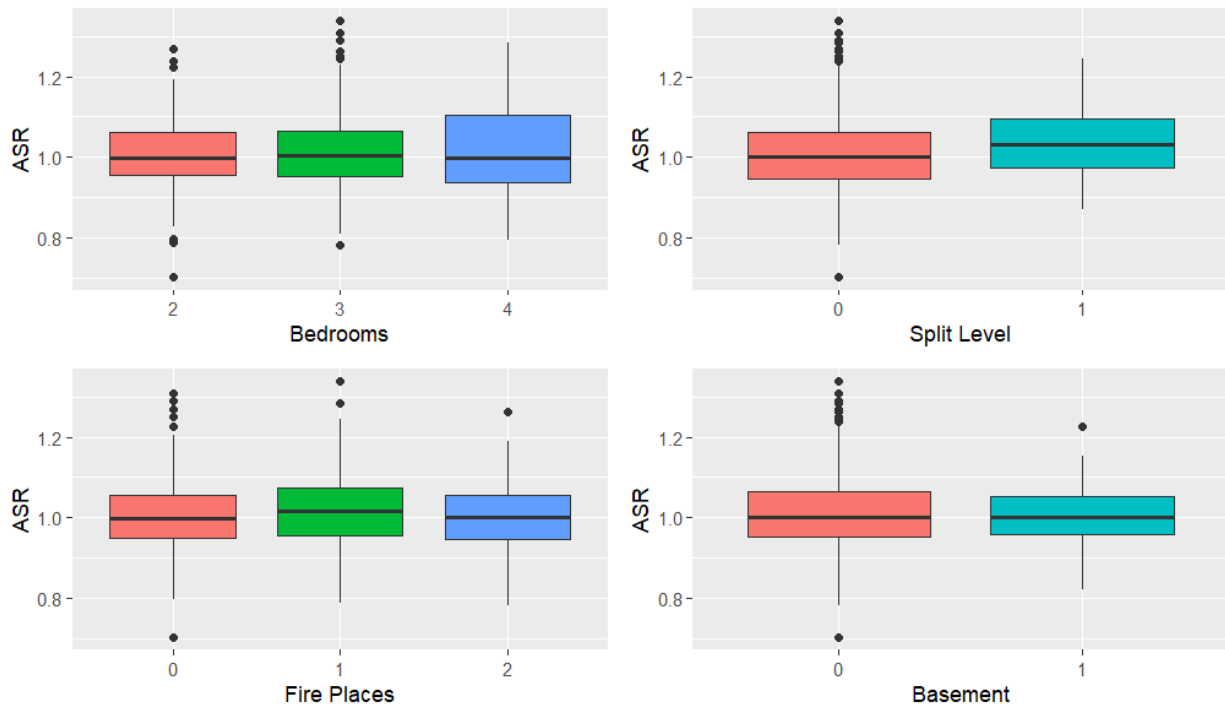
Ratio Statistics for Adjusted ASR (by Neighborhood)

Mean	1.01
95% Confidence Interval for Mean	Lower Bound 1.00
	Upper Bound 1.02
Median	1.00
95% Confidence Interval for Median	Lower Bound 0.99
	Upper Bound 1.01
	Actual Coverage 94.79%
Minimum	0.71
Maximum	1.34
Coefficient of Dispersion	7.32

Which looks very good. Finally, I check if the ASR is equally distributed among various attributes of building and land. Overall, low COD shows there should not be a major vertical inequality in the assessment. First, let's compare the ASR for various living and land areas:



While large buildings and lands seem to be slightly overvalued, the difference does not seem to be significant based on the R^2 . I repeat the same process for the categorical features too by putting box plot



Again, the ASR seems to be equally distributed among the categorical features too.

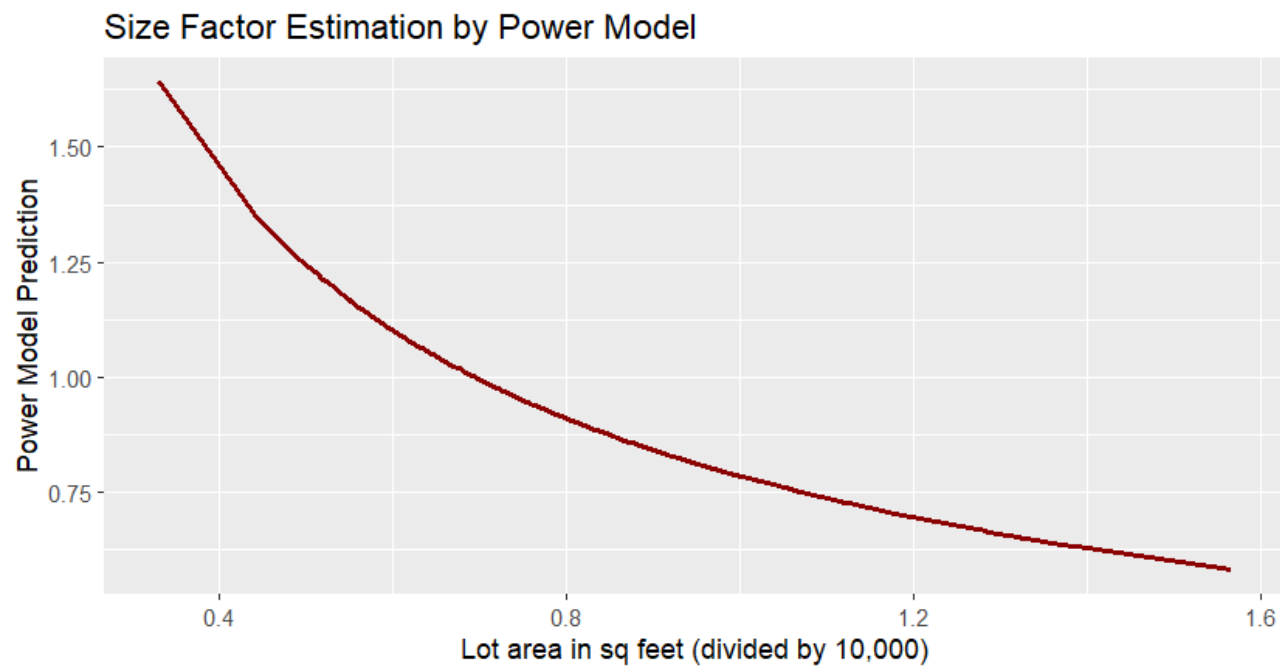
Conclusion

The final cost valuation demonstrates acceptable performance, with a median Assessment-to-Sales Ratio (ASR) of 1 and a Coefficient of Dispersion (COD) of 7.32%, both meeting IAAO standards. Land valuation utilized the unit value method, while improvement valuations were derived from Replacement Cost New (RCN) data, incorporating appreciation and market adjustments. Throughout the process, necessary adjustments were made to account for specific property characteristics and market variations. Statistical tests and figures confirm the model's consistent application across diverse building and land features. Furthermore, the observed minimum and maximum ASR values fall within a reasonable range, signifying the model's effective operation as intended.

Appendix

Calculating Land Residual

Fig A 1 Size Factor Estimation by Power Model



Unit Valuation Method

Fig A 2. Land Residual Ratio after Adjustment for Land Features and Neighborhood

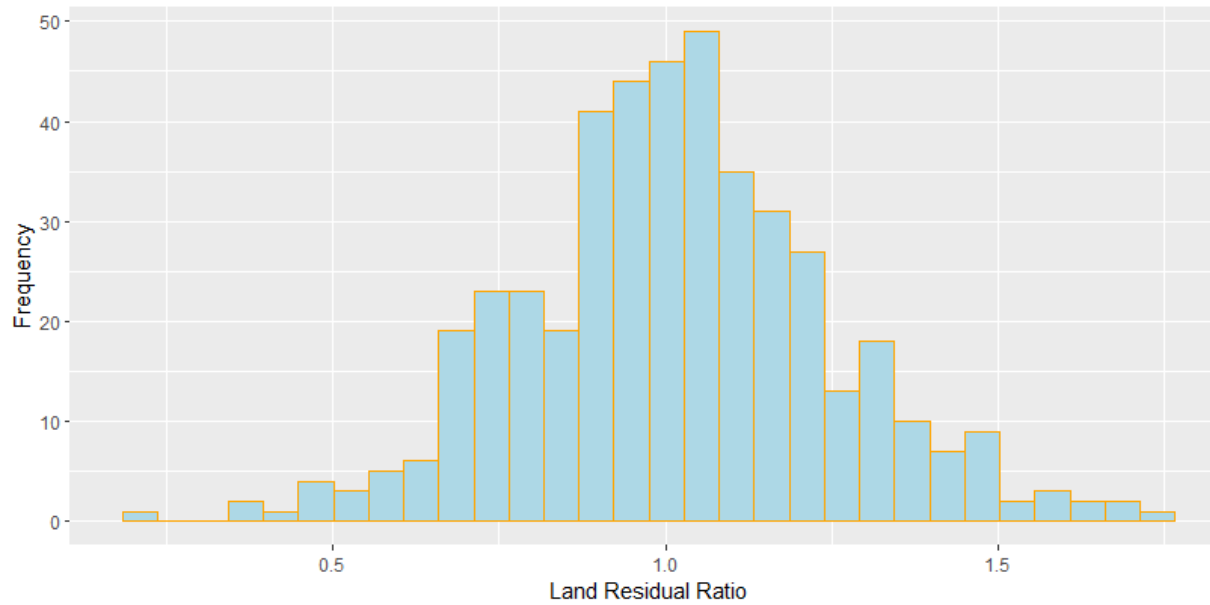


Table A 1 Mann-Whitney Results - general land value adjusted for culdesac

Mann-Whitney Results - general land value adjusted for culdesac				
Variable	Value	Number	Mean Rank	Ays. Significance
corner	0	378	219	0.12
	1	68	246	
culdesac	0	417	225	0.48
	1	29	207	
alley	0	301	247	0.00
	1	145	175	
wooded	0	410	225	0.35

	1	36	204	
pie_shpe	0	420	221	0.12
	1	26	261	
slope	0	389	225	0.53
	1	57	213	
abv_road	0	367	228	0.12
	1	79	203	

Table A 2 **Mann-Whitney Results - general land value adjusted for culdesac and alley**

Mann-Whitney Results - general land value adjusted for culdesac and alley				
Variable	Value	Number	Mean Rank	Ays. Significance
corner	0	378	220	0.19
	1	68	242	
culdesac	0	417	227	0.06
	1	29	179	
alley	0	301	221	0.60
	1	145	228	
wooded	0	410	226	0.11
	1	36	191	
pie_shpe	0	420	223	0.67
	1	26	234	
slope	0	389	226	0.37
	1	57	209	
abv_road	0	367	229	0.04
	1	79	197	

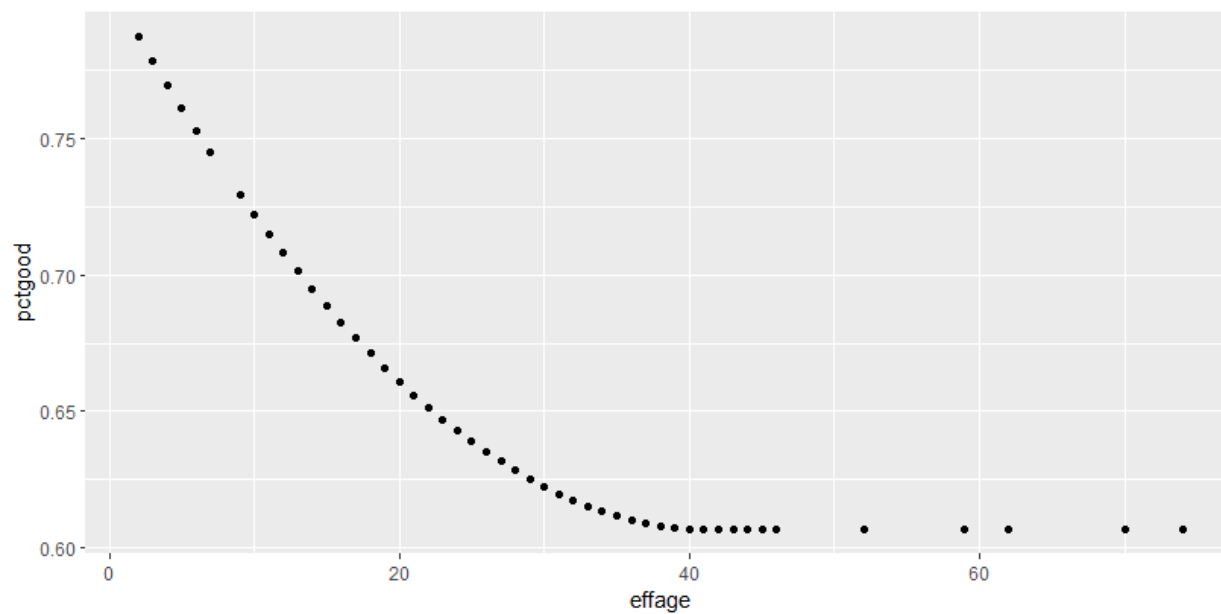
Table A 3 **Mann-Whitney Results - general land value adjusted for culdesac, alley, and abv_road**

Mann-Whitney Results - general land value adjusted for culdesac, alley, and abv_road

Variable	Value	Number	Mean Rank	Ays. Significance
corner	0	378	221	0.29
	1	68	239	
culdesac	0	417	225	0.41
	1	29	205	
alley	0	301	223	0.84
	1	145	225	
wooded	0	410	227	0.09
	1	36	189	
pie_shpe	0	420	222	0.47
	1	26	241	
slope	0	389	223	0.70
	1	57	230	
abv_road	0	367	224	0.87
	1	79	222	

Market-based Depreciation Factor

Fig A 3. Depreciation Factor versus Effective Age



Testing Building Characteristics

Table A 4. Ratio Statistics for Mkfact1 adjusted for Split Level (Mkfactor2)

Split-level	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
0	1.00	0.98	1.02	95.54	0.41	1.79	13.82
1	1.00	0.89	1.08	95.90	0.61	1.21	12.53

Table A 5. Ratio Statistics for Mkfact3 (adjusted by Split-level and Garage Type) based on Bedrooms

Garage Type	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
2	0.95	0.91	0.98	94.51	0.55	1.78	14.22
3	1.02	1	1.04	95.05	0.41	1.55	12.27
4	0.93	0.85	1.02	92.16	0.62	1.53	19.03
2	0.95	0.91	0.98	94.51	0.55	1.78	14.22

Table A 6. Ratio Statistics for Mkfact4 (adjusted by Split-level, Garage Type, and Bedrooms) based on Carport

Garage Type	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
2	0.95	0.91	0.98	94.51	0.55	1.78	14.22
3	1.02	1	1.04	95.05	0.41	1.55	12.27
4	0.93	0.85	1.02	92.16	0.62	1.53	19.03
2	0.95	0.91	0.98	94.51	0.55	1.78	14.22

Table A 7. Ratio Statistics for Mkfact5 (adjusted by Split-level, Garage Type, Bedrooms, and Carport) based on Family Room

Garage Type	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
0	0.95	0.93	0.98	94.33	0.39	1.85	15.16
1	1.03	1.01	1.05	95.09	0.6	1.63	11.59
2	1.01	0.97	1.05	93.61	0.71	1.19	8.15